

ORIGINAL ARTICLE

Urban Spatial Accessibility of Primary Care and Hypertension Control and Awareness on Chicago's South Side: A Study From the COMPASS Cohort

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BACKGROUND: Understanding the relationship between hypertension and spatial accessibility of primary care can inform interventions to improve hypertension control and awareness, especially among disadvantaged populations. This study aims to investigate the association between spatial accessibility of primary care and hypertension control and awareness.

METHODS: Participant data from the COMPASS (Chicago Multiethnic Prevention and Surveillance Study) between 2013 and 2019 were analyzed. All participants were geocoded. Locations of primary care providers in Chicago were obtained from MAPSCorps. A score was generated for spatial accessibility of primary care using an enhanced 2-step floating catchment area method. A higher score indicates greater accessibility. Measured hypertension was defined as systolic blood pressure ≥ 130 mm Hg or diastolic blood pressure ≥ 80 mm Hg. Logistic regression was used to estimate odds ratio and 95% CI for hypertension status in relation to accessibility score quartiles.

RESULTS: Five thousand ninety-six participants (mean age, 53.4 ± 10.8) were included. The study population was predominantly non-Hispanic black (84.0%), over 53% reported an annual household income $< \$15\,000$, and 37.3% were obese. Measured hypertension prevalence was 78.7% in this population, among which 37.7% were uncontrolled and 41.0% were unaware. A higher accessibility score was associated with lower measured hypertension prevalence. In fully adjusted models, compared with the first (lowest) quartile of accessibility score, the odds ratio strengthened from 0.82 (95% CI, 0.67–1.01) for the second quartile to 0.75 (95% CI, 0.62–0.91) for the third quartile, and further to 0.73 (95% CI, 0.60–0.89) for the fourth (highest) quartile. The increasing trend had a $P < 0.01$. Similar associations were observed for both uncontrolled and unaware hypertensions. When stratified by neighborhood socioeconomic status, a higher accessibility score was associated with lower rates of unaware hypertension in both disadvantaged and nondisadvantaged neighborhoods.

CONCLUSIONS: Better spatial accessibility of primary care is associated with improved hypertension awareness and control.

Key Words: blood pressure ■ Chicago ■ hypertension ■ prevalence ■ primary health care

See Editorial by Kershaw

Hypertension is one of the leading preventable factors for disease burden and cardiovascular disease-related mortalities.¹ Community clinics and primary care facilities can detect hypertension and provide affordable medication for hypertension

control, reducing the risk of hypertension and its associated comorbidities and mortality.^{2,3} In 2021, 80 million Americans were living in health professional shortage areas as estimated.⁴ Improving accessibility of primary care is thus prioritized in many initiatives

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Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/CIRCOUTCOMES.121.008845>.

For Sources of Funding and Disclosures, see page 698.

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Circulation: Cardiovascular Quality and Outcomes is available at <http://www.ahajournals.org/journal/circoutcomes>

WHAT IS KNOWN

- Better spatial accessibility of primary care is associated with improved hypertension awareness and control, but whether residents in disadvantaged neighborhoods benefit from enhanced spatial accessibility is unclear.

WHAT THE STUDY ADDS

- In a predominantly lower socioeconomic status cohort, spatial accessibility of primary care is still associated with hypertension awareness and control even after adjustment for or stratification by neighborhood socioeconomic status. This suggests that residents in disadvantaged areas can benefit from interventions that enhance access to primary care physicians.

Nonstandard Abbreviations and Acronyms

ADI	area deprivation index
COMPASS	Chicago Multiethnic Prevention and Surveillance Study
E2FCA	enhanced two-step floating catchment area
OR	odds ratio
SES	socioeconomic status

and interventions for hypertension control and health disparity mitigation. Although most prior studies have evaluated the impacts of primary care accessibility on service utilization or social factors in access inequalities,⁵⁻⁹ specific health outcomes such as hypertension were generally overlooked.⁵

As one of the most racially segregated cities in the United States,¹⁰ Chicago has seen a widening gap in health disparities over the past decades.¹¹ On the Chicago's South Side, the largest continuous Black urban community in the United States, longstanding structural racism in the housing system has led to substantial challenges including concentrated poverty, violence, and food deserts.^{12,13} A 30-year gap in life expectancy has been observed between South Side neighborhoods and Chicago's wealthier northern neighborhoods,¹⁴ which could be largely attributable to high rates of hypertension, heart disease, and stroke.¹⁵ A shortage of medical resources that prevents South Side residents from accessing primary care contributes to these alarming racial inequalities in health.^{16,17}

The COMPASS (Chicago Multiethnic Prevention and Surveillance Study), a population-based longitudinal study, was initiated in 2013 to study health disparities among the population of Chicago's South Side. Given the complex urban settings and health inequalities

experienced by Black people in Chicago, COMPASS provides a unique opportunity to investigate the association between spatial accessibility of primary care and hypertension awareness and control. As prior studies suggest that neighborhood socioeconomic status (SES) plays a role in spatial accessibility of primary care and its associated impacts, we incorporated neighborhood SES in our analysis and evaluated its effect modifying role.

METHOD

Study Population

Details of COMPASS have been described elsewhere.¹² Residents of the greater Chicago area are eligible for COMPASS if they are (1) age 35 or older at the time of enrollment; (2) able to give consent and provide survey data in English or Spanish; (3) willing to provide blood, urine, and saliva samples as well as electronic health record access. COMPASS has employed multiple recruitment modalities, including a population-based approach, a community-based approach, and a hospital/clinic-based approach. During their first visit to the research clinic, participants complete the survey questionnaire. Anthropometry and blood pressure are then ascertained. COMPASS questionnaire codebooks can be found at <https://compass.uchicago.edu/research/self-reported-questionnaire-data/>. The data that support the findings of this study are available from the corresponding author upon reasonable request.

As of 2019, 7728 participants have been recruited. Their residential addresses were collected at enrollment and have been verified and geocoded by research staff. The University of Chicago Institutional Review Board granted approval for this study and written informed consent was obtained from all participants.

Hypertension

We defined 2 types of hypertension: self-reported hypertension and measured hypertension. Measured hypertension was further classified into uncontrolled hypertension and unaware hypertension. Among participants with self-reported hypertension, we identified hypertension medication users and nonusers.

Self-reported hypertension was measured in the survey, which asked "Has a doctor ever told you that you have hypertension?" Participants who answered "yes" were further asked, "Do you currently take medicine to treat your high blood pressure (hypertension)?"

Systolic and diastolic blood pressure was measured by trained staff using an Omron HEM 907XL blood pressure monitor. Staff were trained to allow participants to rest in a sitting position for five minutes before measurement and to ask participants to sit upright with both feet flat on the floor during measurement. Average systolic blood pressure and diastolic blood pressure were generated from 2 measures.

This study follows the 2017 American College of Cardiology/American Heart Association Hypertension guidelines for hypertension classification.¹⁸ Specifically, participants would be classified as "measured hypertension" if they had an average systolic blood pressure ≥ 130 mmHg or an average diastolic blood pressure ≥ 80 mmHg; otherwise would be

classified as “no measured hypertension.” “Measured hypertension” was further classified as “uncontrolled hypertension” if the participant self-reported hypertension, and “unaware hypertension” if the participant did not self-report hypertension.

Primary Care Provider Locations

Primary care provider locations were obtained through MAPSCorps, an independent nonprofit organization based on Chicago's South Side.¹⁹ MAPSCorps conducts an annual block-by-block census of community assets that generates public-use data for Chicago, especially the South Side. More details can be found at their website: www.mapscorps.org.

We selected active primary care providers from MAPSCorps data outlining healthcare service provider locations between 2013 and 2019 during COMPASS recruitment. The primary care providers in this study include family physicians, general practitioners, and general internists. As MAPSCorps only covered 57 of 77 communities in Chicago as of 2019, we restricted our study populations to those who lived in these areas.

Covariates

In this study, neighborhood SES was assessed using the Area Deprivation Index (ADI) between 2015 and 2019.²⁰ The ADI measures neighborhood disadvantages and is composed of 17 indices covering education, employment, housing quality, and poverty.²¹ ADI rankings are available online.²² A higher ADI rank indicates more neighborhood disadvantages.

Demographic covariates derived from the COMPASS survey included age (<50, 50–65, >65), race (non-Hispanic White, non-Hispanic Black, Hispanic, other), gender (female, male), body mass index (<25, 25–29.9, 30–39.9, ≥40), education (<high school, high school, some college, college or more), household income (<\$15 000, \$15 000–24 999, \$25 000–34 999, >\$35 000), smoking status (nonsmoker, former smoker, current smoker), and insurance status (not insured, Medicare, Medicaid, private, other).

Spatial Accessibility Score

We mapped participants' residential addresses and locations of primary care providers. Then we measured the spatial accessibility of primary care for each participant using the enhanced 2-step floating catchment area (E2FCA) method, a widely used method for spatial accessibility measure.²³ The E2FCA method measures spatial accessibility in two steps and generates a score for spatial accessibility. In the first step, the E2FCA method evaluates the catchment area of each primary care provider, estimates the overall population in the catchment area, and then generates the provider-to-population ratio for each provider. In the second step, for each participant, the E2FCA method identifies the provider with the catchment area within which the participant lives and sums up the provider-to-population ratios of all these providers. A distance decay effect was assumed in both steps that an increase in distance would lead to less service utilization. Therefore, the accessibility score is essentially the overall provider-to-population ratio at the individual level. A higher score indicates better accessibility. More details about E2FCA can be found in the [Supplemental Material](#).^{23–26}

Statistical Analysis

We used logistic regression to estimate the odds ratio (OR) and 95% CI for self-reported hypertension, hypertension medication use, and measured hypertension according to quartiles of the accessibility score. We generated the median value of each accessibility score quartile and treated it as a continuous variable in the model to calculate the *P* values for trend.

We constructed 3 logistic regression models. The first model was the unadjusted model. The second model was the directed acyclic graph (DAG) based model. We constructed a DAG ([Figure S1](#)) and selected confounding variables accordingly. As a result, age, race, gender, education, and household income were selected and adjusted. The third model was the fully adjusted model. We adjusted for all available potential confounders to control for potential residual confounding. In addition to covariates adjusted in the second model, we further adjusted for insurance status, body mass index, smoking status, and ADI (quadratic term) in the model.

To investigate the effect modification of neighborhood SES,^{27–29} we stratified our study populations into two groups based on ADI national rank: ≤50th and >50th percentiles. We estimated the OR and 95% CI in each stratum, and the *P* value for interaction by adding a cross-product term between accessibility score and ADI group in the third fully adjusted model.

To evaluate potential nonlinear relations, we fitted a generalized additive model including a spline term for the accessibility score with 3 *df* and visualized the exposure–outcome response. We used the 10th percentile of accessibility score (44.1 per 100 000) as the reference and estimated the odds ratio compared to the reference.

Multiple imputation was used for missing values in covariates in the regression models. We assumed they were missing at random and imputed the missing values using fully conditional specification.³⁰ All covariates and the accessibility score were used to generate the imputed data. Ten complete datasets were generated via imputation, and we employed the analytical procedure PROC MIANALYZE in SAS to combine the results.³¹

In sensitivity analysis, we evaluated the influence of changes in hypertension guidelines in the fully adjusted model. We reclassified measured hypertension according to guidelines before 2017 (ie, hypertension was defined as an average systolic blood pressure ≥140 mm Hg or an average diastolic blood pressure ≥90 mm Hg).

Spatial analysis was conducted using ArcGIS Pro 2.6.0 (Esri, Inc, West Redlands, CA). All other statistical analyses were performed using SAS version 9.4 (SAS Institute, Inc, Cary, NC). All statistical tests were 2-tailed, and *P* < 0.05 were considered statistically significant.

RESULTS

Among the source population of 7728 participants, 5921 had valid data on blood pressure. Furthermore, 5096 of them resided in communities covered by MAPSCorps. Therefore, these 5096 participants were the final study population. The study population was predominantly non-Hispanic black (84.0%), over 53% reported an annual household income <\$15 000, 37.3% were obese, and 53.5% were current smokers (Table 1).

Table 1. Distributions of Selected Demographic Characteristics in the Study Population and Across the Quartiles of Spatial Accessibility Scores

Demographic characteristics	Overall (n=5096)	The first quartile of accessibility score (n=1274)	The second quartile of accessibility score (n=1273)	The third quartile of accessibility score (n=1275)	The fourth quartile of accessibility score (n=1274)
Blood pressure (mean [SD])					
Systolic	136.2 (21.9)	136.6 (23.2)	137.3 (21.5)	135.7 (21.3)	135.2 (21.7)
Diastolic	86.4 (14.1)	87.8 (15.1)	86.4 (14.2)	85.2 (13.0)	86.2 (13.7)
Self-report hypertension					
No	2833 (56.1)	690 (54.2)	738 (58.0)	756 (59.3)	673 (52.8)
Yes	2220 (43.9)	584 (45.8)	535 (42.0)	519 (40.7)	601 (47.2)
Hypertension medication use	1835 (36.0)	465 (36.5)	443 (34.8)	434 (34.0)	493 (38.7)
Measured hypertension					
No	1083 (21.3)	241 (18.9)	259 (20.4)	290 (22.8)	293 (23.0)
Yes	4013 (78.7)	1033 (81.1)	1014 (79.7)	985 (77.3)	981 (77.0)
Uncontrolled	1921 (37.7)	510 (40.0)	457 (35.9)	443 (34.8)	511 (40.1)
Unaware	2092 (41.0)	523 (41.1)	557 (43.8)	542 (42.5)	470 (36.9)
Age					
<50	1869 (37.0)	501 (39.3)	477 (37.5)	444 (34.8)	463 (36.3)
50–65	2528 (50.0)	607 (47.7)	636 (50.0)	682 (53.5)	629 (49.4)
>65	652 (12.9)	165 (13.0)	160 (12.6)	146 (11.5)	182 (14.3)
Missing	4 (0.1)	1 (0.1)	0 (0)	3 (0.2)	0 (0)
Race					
White	309 (6.1)	138 (10.8)	54 (4.2)	31 (2.4)	87 (6.8)
Black	4246 (84.0)	952 (74.7)	1156 (90.8)	1130 (88.6)	1044 (82.0)
Hispanic	362 (7.2)	150 (11.8)	24 (1.9)	79 (6.2)	112 (8.8)
Other	59 (1.2)	17 (1.3)	16 (1.3)	14 (1.1)	13 (1.0)
Missing	77 (1.5)	17 (1.3)	23 (1.8)	21 (1.7)	18 (1.4)
Gender					
Female	2548 (50.4)	673 (52.8)	649 (51.0)	597 (46.8)	646 (50.7)
Male	2495 (49.4)	601 (47.2)	621 (48.8)	673 (52.8)	626 (49.1)
Missing	10 (0.2)	0 (0)	3 (0.2)	5 (0.4)	2 (0.2)
BMI					
<25	1597 (31.6)	394 (30.9)	384 (30.2)	431 (33.8)	402 (31.6)
25–29.9	1427 (28.2)	349 (27.4)	368 (28.9)	331 (26.0)	387 (30.4)
30–39.9	1483 (29.4)	402 (31.6)	386 (30.3)	348 (27.3)	362 (28.4)
≥40	398 (7.9)	116 (9.1)	99 (7.8)	93 (7.3)	95 (7.5)
Missing	148 (2.9)	13 (1.0)	36 (2.8)	72 (5.7)	28 (2.2)
Education					
<High school	1715 (33.9)	391 (30.7)	372 (29.2)	468 (36.7)	499 (39.2)
High school	1423 (28.2)	356 (27.9)	365 (28.7)	376 (29.5)	338 (26.5)
Some college	1486 (29.4)	383 (30.1)	410 (32.2)	369 (28.9)	337 (26.5)
College or more	404 (8.0)	141 (11.1)	120 (9.4)	51 (4.0)	95 (7.5)
Missing	25 (0.5)	3 (0.2)	6 (0.5)	11 (0.9)	5 (0.4)
Household income					
<\$15 000	2679 (53.0)	644 (50.6)	630 (49.5)	699 (54.8)	732 (57.5)
\$15 000–\$24 999	812 (16.1)	202 (15.9)	240 (18.9)	174 (13.7)	202 (15.9)
\$25 000–\$34 999	278 (5.5)	81 (6.4)	70 (5.5)	54 (4.2)	76 (6.0)
>\$34 999	619 (12.3)	219 (17.2)	185 (14.5)	73 (5.7)	147 (11.5)
Missing	665 (13.2)	128 (10.1)	148 (11.6)	275 (21.6)	117 (9.2)
Smoking status					
Nonsmoking	1463 (29.0)	404 (31.7)	387 (30.4)	315 (24.7)	371 (29.1)

(Continued)

Table 1. Continued

Demographic characteristics	Overall (n=5096)	The first quartile of accessibility score (n=1274)	The second quartile of accessibility score (n=1273)	The third quartile of accessibility score (n=1275)	The fourth quartile of accessibility score (n=1274)
Former smoker	668 (13.2)	214 (16.8)	150 (11.8)	124 (9.7)	182 (14.3)
Current smoker	2701 (53.5)	631 (49.5)	687 (54.0)	749 (58.8)	657 (51.6)
Missing	221 (4.4)	25 (2.0)	49 (3.9)	87 (6.8)	64 (5.0)
Insurance status					
No	599 (11.8)	212 (16.6)	126 (9.9)	82 (6.4)	179 (14.1)
Medicaid	1716 (33.7)	379 (29.8)	390 (30.6)	536 (42.0)	411 (32.3)
Medicare	738 (14.5)	163 (12.8)	215 (16.9)	198 (15.5)	162 (12.7)
Private	705 (13.8)	235 (18.5)	185 (14.5)	102 (8.0)	183 (14.4)
Other	1332 (26.2)	283 (22.2)	354 (27.8)	356 (27.9)	339 (26.6)
Missing	6 (0.1)	2 (0.2)	3 (0.2)	1 (0.1)	0 (0)
Area deprivation index national percentile					
Median (interquartile range)	71 (52–86)	67 (51–87)	71 (61–79)	73 (57–75)	69 (42–87)
Spatial accessibility score					
Range	16.4–88.9	16.4–51.0	51.0–70.1	70.1–80.3	80.3–88.9

BMI indicates body mass index; and SA, spatial accessibility.

Most of our study population resided in a neighborhood with an ADI rank larger than 70th percentile. The geographic distributions of study population, primary care providers, and ADI rank, as well as accessibility score quartiles, are shown in Figure 1. The catchment area for each primary care provider is presented in Figure S2. The accessibility score ranged from 16.4 to 86.6 per 100 000, with a median value of 51.0 per 100 000. Characteristics related to primary care, such as number of primary care providers within 10min drive, were presented in Table S1.

Fewer than half (43.9%) of participants reported hypertension, among whom 82.0% reported current use of hypertension medication. The prevalence rate was higher for measured hypertension (78.7% according to 2017 American College of Cardiology/American Heart Association guidelines or 53.6% according to earlier guidelines [Table S2]). Among the 78.7% with measured hypertension, 37.7% were aware but uncontrolled, while the other 41.0% were unaware. The hypertension control rate was less than 15% (or less than 35% according to earlier guidelines). The community hypertension prevalence in the study population was higher than that reported in the Chicago Health Atlas (Figure S3).

We observed significant associations between spatial accessibility of primary care and hypertension in the study population, especially in the fully adjusted model (Table 2). In the unadjusted model, no significant association was observed between self-reported hypertension and accessibility score. However, when adjusted for potential confounders, the associations were strengthened in the DAG-based model and the fully adjusted model. In the fully adjusted model, self-reported hypertension exhibited nonlinear associations with accessibility score. Specifically, a reduced prevalence was observed

for self-reported hypertension in the second (OR, 0.83 [95% CI, 0.69–0.99]) and third (OR, 0.80 [95% CI, 0.67–0.95]) quartiles of accessibility score. Then the association was attenuated, and no significant association was observed for the fourth quartile. However, spatial accessibility of primary care was not associated with hypertension medication use among participants who self-reported hypertension in any models.

For measured hypertension, the results were similar between unadjusted and DAG-adjusted models, whereas the associations were further strengthened in the fully adjusted model. Overall, the results were consistent across the 3 models. In the fully adjusted model, the magnitude of association for measured hypertension increased as the accessibility score increased (P for trend: <0.01), with significant associations observed for the third (OR, 0.75 [95% CI, 0.62–0.91]) and fourth (OR, 0.73 [95% CI, 0.60–0.89]) quartiles.

When looking into the subclasses of measured hypertension, different patterns were observed for uncontrolled hypertension and unaware hypertension. In the fully adjusted model, compared to the lowest accessibility score quartile, a reduced prevalence of uncontrolled hypertension was observed for the third quartile in all models (eg, the fully adjusted model: OR, 0.60 [95% CI, 0.47–0.76]). Significant associations with the second and the fourth quartiles were only observed in the fully adjusted model for uncontrolled hypertension. On the other hand, unaware hypertension exhibited a more linear relationship with accessibility scores. In the fully adjusted model, compared to the lowest quartile, the ORs for unaware hypertension strengthened from 0.91 (95% CI, 0.73–1.14) for the second quartile to 0.81 (95% CI, 0.65–1.01) for the third quartile and further to 0.71 (95% CI, 0.57–0.88) for the fourth quartile (P for trend: <0.01).

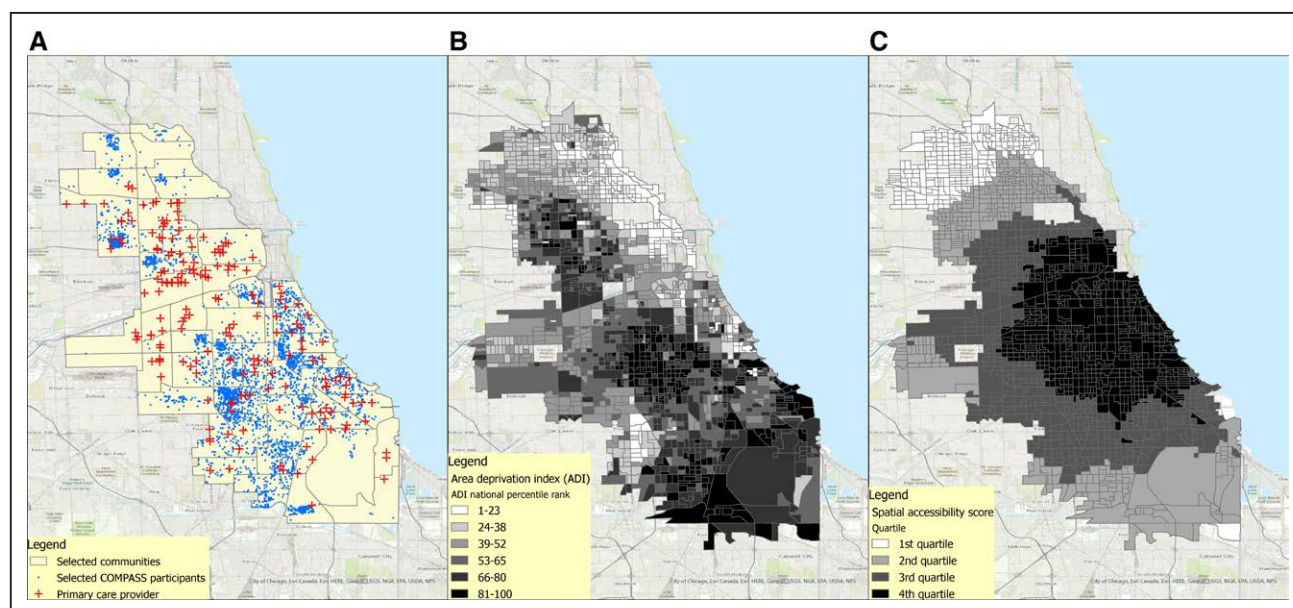


Figure 1. Descriptive maps for study population from the COMPASS (Chicago Multiethnic Prevention and Surveillance Study) between 2013 and 2019.

(A) geographic distribution of study population and primary care providers; (B) distribution of area deprivation index (ADI) national percentile rank in selected communities; (C) distribution of spatial accessibility scores.

Similar trends were also observed in unadjusted and DAG-based models.

When stratified by neighborhood SES (ADI ranks ≤ 50 th versus > 50 th percentiles) in the fully adjusted model, a few differences were observed (Table 3). Compared with the lowest quartile of accessibility scores, the highest quartile was significantly associated with lower hypertension prevalence in both strata; however, the association was more pronounced in less disadvantaged neighborhoods (ADI rank ≤ 50 th percentile: OR, 0.54 [95% CI, 0.35–0.83] versus ADI rank > 50 th percentile: OR, 0.76 [95% CI, 0.58–0.99]). Moreover, higher score quartiles were significantly associated with lower prevalence of uncontrolled hypertension only in neighborhoods with ADI ranks > 50 th percentile (eg, second quartile, OR, 0.68 [95% CI, 0.52–0.88]), but not in neighborhoods with ADI ≤ 50 th percentile (eg, second quartile, OR, 0.67 [95% CI, 0.35–1.31]). Nonetheless, it should be noted that none of these differences were statistically significant.

Results from the spline model were generally consistent with those from logistic models (Figure 2). Nonlinear association was observed between accessibility score and self-reported hypertension. The odds for self-reported hypertension increased at the lower levels of accessibility score, followed by a U shape. Higher accessibility score appeared to be associated with higher odds for hypertension medication use, but the association was not significant across the whole range. By contrast, higher score was associated with lower odds for measured hypertension. However, for uncontrolled hypertension, the inverse association

was attenuated when the accessibility score was high, whereas the decreasing trend was consistent for unaware hypertension.

In sensitivity analysis using hypertension guidelines before 2017, the results were consistent with our main analysis (Tables S3 and S4). Across the study population, higher accessibility scores were significantly associated with reduced prevalence for both uncontrolled and unaware hypertensions, and these associations appeared to be stronger as accessibility scores increased. The results also remained similar when stratified by ADI national rank. Overall, using guidelines before 2017 did not substantially change our results.

DISCUSSION

In this study, we investigated the association between spatial accessibility of primary care and hypertension control and awareness using participants from COMPASS. The study population mainly consists of urban non-Hispanic blacks and low-income families. We observed that better spatial accessibility of primary care was associated with lower hypertension prevalence including both uncontrolled and unaware hypertension, but not with hypertension medication use among participants who self-reported hypertension. When stratified by neighborhood SES, better spatial accessibility was associated with lower prevalence of unaware hypertension in both disadvantaged and nondisadvantaged neighborhoods, although this association was more pronounced in nondisadvantaged neighborhoods. Spatial accessibility was associated with uncontrolled hypertension only

Table 2. Adjusted OR and 95% CI for Hypertension Status According to Spatial Accessibility Score Quartiles

	Self-reported hypertension				Measured hypertension				Uncontrolled hypertension vs No measured hypertension			
	Yes vs no		Medication use vs no medication use		All measured hypertension vs No measured hypertension		Uncontrolled hypertension vs No measured hypertension		Unaware hypertension vs No measured hypertension			
	N*	OR (95% CI)	N*	OR (95% CI)	N*	OR (95% CI)	N*	OR (95% CI)	N*	OR (95% CI)	N*	OR (95% CI)
Model 1: unadjusted model†												
First (lowest) quartile (16.4–51.0 per 100 000)	584/690	Ref	461/123	Ref	1033/241	Ref	510/241	Ref	523/241	Ref	523/241	Ref
Second quartile (51.0–70.1 per 100 000)	535/738	0.95 (0.81–1.11)	431/104	1.11 (0.83–1.48)	1014/259	0.91 (0.75–1.11)	457/259	0.83 (0.67–1.04)	557/259	0.99 (0.80–1.23)	557/259	0.99 (0.80–1.23)
Third quartile (70.1–80.3 per 100 000)	519/756	0.90 (0.77–1.05)	430/89	1.29 (0.95–1.75)	985/290	0.79 (0.65–0.96)	443/290	0.72 (0.58–0.89)	542/290	0.86 (0.70–1.06)	542/290	0.86 (0.70–1.06)
Fourth (highest) quartile (80.3–88.9 per 100 000)	601/673	1.07 (0.92–1.25)	476/125	1.02 (0.77–1.34)	981/293	0.78 (0.65–0.95)	511/293	0.82 (0.67–1.02)	470/293	0.74 (0.60–0.91)	470/293	0.74 (0.60–0.91)
P for trend		0.77		0.52		<0.01		0.02		<0.01		<0.01
Model 2: DAG-based model‡												
First (lowest) quartile (16.4–51.0 per 100 000)	584/690	Ref	461/123	Ref	1033/241	Ref	510/241	Ref	523/241	Ref	523/241	Ref
Second quartile (51.0–70.1 per 100 000)	535/738	0.86 (0.73–1.02)	431/104	1.11 (0.83–1.48)	1014/259	0.90 (0.74–1.10)	457/259	0.82 (0.66–1.02)	557/259	0.96 (0.78–1.19)	557/259	0.96 (0.78–1.19)
Third quartile (70.1–80.3 per 100 000)	519/756	0.84 (0.71–1.00)	430/89	1.29 (0.96–1.75)	985/290	0.80 (0.66–0.97)	443/290	0.72 (0.58–0.89)	542/290	0.83 (0.67–1.03)	542/290	0.83 (0.67–1.03)
Fourth (highest) quartile (80.3–88.9 per 100 000)	601/673	0.97 (0.82–1.15)	476/125	1.03 (0.78–1.36)	981/293	0.78 (0.64–0.95)	511/293	0.85 (0.69–1.06)	470/293	0.74 (0.60–0.91)	470/293	0.74 (0.60–0.91)
P for trend		0.48		0.47		<0.01		0.03		<0.01		<0.01
Model 3: fully adjusted model§												
First (lowest) quartile (16.4–51.0 per 100 000)	584/690	Ref	461/123	Ref	1033/241	Ref	510/241	Ref	523/241	Ref	523/241	Ref
Second quartile (51.0–70.1 per 100 000)	535/738	0.82 (0.69–0.99)	431/104	0.99 (0.72–1.36)	1014/259	0.82 (0.67–1.01)	457/259	0.71 (0.55–0.90)	557/259	0.91 (0.73–1.14)	557/259	0.91 (0.73–1.14)
Third quartile (70.1–80.3 per 100 000)	519/756	0.79 (0.66–0.95)	430/89	1.24 (0.90–1.72)	985/290	0.75 (0.62–0.91)	443/290	0.60 (0.47–0.76)	542/290	0.81 (0.65–1.01)	542/290	0.81 (0.65–1.01)
Fourth (highest) quartile (80.3–88.9 per 100 000)	601/673	0.97 (0.82–1.16)	476/125	1.04 (0.77–1.39)	981/293	0.73 (0.60–0.89)	511/293	0.77 (0.62–0.97)	470/293	0.71 (0.57–0.88)	470/293	0.71 (0.57–0.88)
P for trend		0.46		0.48		<0.01		<0.01		<0.01		<0.01

BMI indicates body mass index; DAG, directed acyclic graph; and OR, odds ratio.

*The number of cases/controls.

†The models were not adjusted for any confounders.

‡The models were adjusted for confounders selected from DAG, including age (<50, 50–65, >65), race (non-Hispanic white, non-Hispanic black, Hispanic, other), gender (female, male), education (< high school, high school, some college, college or more), and household income (<\$15 000, \$15 000–\$24 999, \$25 000–\$34 999, >\$35 000).

§The models were adjusted for all potential confounders in this study. In addition to confounders adjusted in the DAG-based model, the fully adjusted model further adjusted for body mass index (BMI; <25, 25–29.9, 30–39.9, ≥40), smoking status (nonsmoker, former smoker, current smoker), insurance status (not insured, Medicaid, Medicare, private, and other), and area deprivation index (quadratic term).

Table 3. OR and 95% CI for Hypertension Status According to Spatial Accessibility Score Quartiles in Populations Stratified by Area Deprivation Index in the Fully Adjusted Model

	Self-reported hypertension			Measured hypertension			Uncontrolled hypertension vs no measured hypertension			Unaware hypertension vs no measured hypertension			
	Yes vs no		Medication use vs no medication use	All measured hypertension vs no measured hypertension		Uncontrolled hypertension vs no measured hypertension	Unaware hypertension vs no measured hypertension						
	N*	OR (95% CI)		N*	OR (95% CI)		N*	OR (95% CI)					
Area deprivation index rank ≤50th percentile													
First (lowest) quartile (16.4–51.0 per 100 000)	121/205	Ref	90/31	Ref	249/77	Ref	108/77	Ref	141/77	Ref			
Second quartile (51.0–70.1 per 100 000)	48/83	0.78 (0.45–1.34)	44/4	3.44 (0.96–12.33)	104/27	0.76 (0.44–1.32)	45/27	0.67 (0.35–1.31)	59/27	0.84 (0.46–1.53)			
Third quartile (70.1–80.3 per 100 000)	115/139	1.26 (0.84–1.89)	96/19	1.94 (0.87–4.34)	198/56	0.74 (0.46–1.20)	101/56	0.84 (0.47–1.49)	97/56	0.69 (0.40–1.17)			
Fourth (highest) quartile (80.3–88.9 per 100 000)	237/253	1.31 (0.90–1.91)	185/52	1.40 (0.74–2.64)	360/130	0.65 (0.44–0.94)	197/130	0.84 (0.53–1.32)	163/130	0.54 (0.35–0.83)			
P for trend		0.06		0.44		0.02		0.56		<0.01			
Area deprivation index rank >50th percentile													
First (lowest) quartile (16.4–51.0 per 100 000)	463/485	Ref	371/92	Ref	784/164	Ref	402/164	Ref	382/164	Ref			
Second quartile (51.0–70.1 per 100 000)	487/655	0.79 (0.66–0.96)	397/100	0.88 (0.63–1.23)	910/232	0.81 (0.65–1.02)	412/232	0.68 (0.52–0.88)	498/232	0.90 (0.71–1.16)			
Third quartile (70.1–80.3 per 100 000)	404/617	0.69 (0.57–0.85)	334/70	1.16 (0.81–1.66)	787/234	0.78 (0.61–0.99)	342/234	0.55 (0.42–0.72)	445/234	0.82 (0.64–1.05)			
Fourth (highest) quartile (80.3–88.9 per 100 000)	364/420	0.91 (0.74–1.12)	291/73	1.02 (0.71–1.45)	621/163	0.71 (0.57–0.89)	314/163	0.75 (0.57–1.00)	307/163	0.76 (0.58–0.99)			
P for trend		0.10		0.61		0.02		<0.01		0.03			
P value for interaction													
First (lowest) quartile (16.4–51.0 per 100 000)		Ref		Ref		Ref		Ref		Ref			
Second quartile (51.0–70.1 per 100 000)		0.61		0.11		0.90		0.91		0.84			
Third quartile (70.1–80.3 per 100 000)		0.46		0.33		0.87		0.26		0.56			
Fourth (highest) quartile (80.3–88.9 per 100 000)		0.28		0.48		0.48		0.83		0.26			

All models were adjusted for age (<50, 50–65, >65), race (non-Hispanic white, non-Hispanic black, Hispanic, other), gender (female, male), body mass index (BMI; <25, 25–29.9, 30–39.9, ≥40), education (<high school, high school, some college, college or more), household income (<\$15 000, \$15 000–24 999, \$25 000–34 999, ≥\$35 000), smoking status (nonsmoker, former smoker, current smoker), and insurance status (not insured, Medicaid, Medicare, private, and other). BMI indicates body mass index; and OR, odds ratio.

* The number of cases/controls.

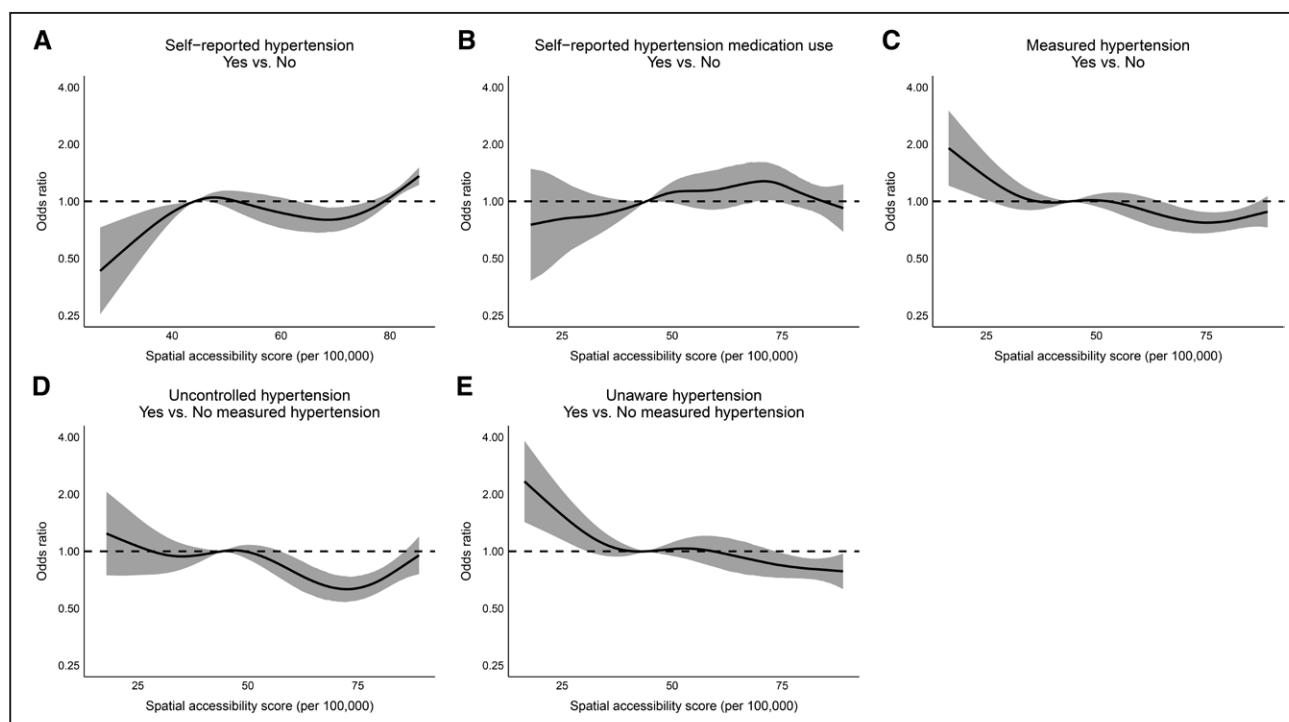


Figure 2. Odds ratio for hypertension status according to spatial accessibility score using a generalized additive model (GAM) with a spline term and 3 df:

A, self-reported hypertension; **(B)** self-reported hypertension medication use; **(C)** measured hypertension; **(D)** uncontrolled hypertension; and **(E)** unaware hypertension. The 10th percentile of spatial accessibility score was used as the reference. The gray area represents 95% CIs. All models adjusted for age (<50, 50–65, >65), race (non-Hispanic white, non-Hispanic black, Hispanic, other), gender (female, male), body mass index (<25, 25–29.9, 30–39.9, ≥40), education (<high school, high school, some college, college or more), household income (<\$15 000, \$15 000–24 999, \$25 000–34 999, >\$35 000), smoking status (nonsmoker, former smoker, current smoker), and insurance status (not insured, Medicaid, Medicare, private, and other).

in disadvantaged neighborhoods. Overall, neighborhood SES may play a role in the association between spatial accessibility of primary care and hypertension prevalence, but it did not eliminate the association of spatial accessibility of primary care in disadvantaged neighborhoods.

The prevalence of hypertension observed in this study is higher than official surveillance data. This difference may be explained by distinct sources for hypertension prevalence. The surveillance data in the Chicago Health Atlas rely on self-report,³² whereas our study defined hypertension using measured blood pressure. Additionally, the oversampling of a disadvantaged population in this study may also contribute to the higher observed hypertension prevalence, as low SES is associated with higher hypertension prevalence.³³ Overall, the higher prevalence in this study suggests an underestimation of hypertension burden in the disadvantaged population on Chicago's South Side.

The low hypertension control rate (15%–35%, depending on which guidelines were used) in this study is worrisome. More than 50% hypertension cases in the United States are under control,³⁴ but it seems that this urban minority population does not benefit from this high control rate. Studies have revealed a lower control rate among Blacks because of multifaceted barriers including insufficient financial support and unsuccessful behavior change.³⁵

The association between self-reported hypertension and accessibility score is interesting. The inverse association in the middle level of accessibility score might imply a consequence of complicated interactions between social factors. On one hand, people in these areas with middle-level scores had better access to medical resources and thus the hypertension prevalence might be lower compared with those in areas with low level of scores. On the other hand, medical resources in these areas were not sufficient enough to serve everyone compared to areas with high accessibility score. However, chance finding cannot be ruled out as well.

We examined the effect modification of neighborhood SES and found a few nonsignificant differences between disadvantaged and nondisadvantaged neighborhoods. These nonsignificant differences suggest a role of neighborhood SES in the associations. Moreover, the beneficial association of better spatial accessibility of primary care did not fade when adjusted for neighborhood SES or restricted to disadvantaged neighborhoods. These findings are contrary to the only prior study on spatial accessibility of primary care and hypertension control in the United States. This study used the MESA (Multi-Ethnic Study of Atherosclerosis) cohort and observed null association between spatial accessibility of primary care and hypertension prevalence when adjusted for demographic and neighborhood characteristics. It thus argues that improving spatial accessibility

of primary care in disadvantaged neighborhoods might not help hypertension control.²⁷ However, this study defined spatial accessibility of primary care as residence in a health professional shortage area, which weakened its conclusion.²⁷ Although residence provides an indication of spatial accessibility, it fails to capture the individual variations in spatial access to primary care. With a better definition at the individual level, results from our study can provide more evidence for the impact of primary care on individual hypertension control and awareness.

Chicago's South Side suffers from substantial health burdens of cardiovascular disease exacerbated by structural racism. Structural racism in Chicago is rooted in 20th century neighborhood redlining, community disinvestment, and racial animus that made homeownership largely unattainable for Blacks, hindering their upward mobility and funneling them to neighborhoods characterized by lower quality housing, infrastructure, and opportunities.³⁶ The multidimensional experience of racism is linked to poorer health care access and outcomes.^{37,38} But the great burden of cardiovascular disease mortality is not inevitable. Studies have shown that higher spatial accessibility to primary care was associated with lower odds for preventable emergency visit in Chicago area,³⁹ which was consistent with our conclusion regarding uncontrolled and unaware hypertension. In the Community Intervention to Reduce Cardiovascular Disease in Chicago (CIRCL-Chicago), an intervention initiated in 2020, researchers aimed to mitigate cardiovascular disease burdens by improving healthcare access among residents on the South Side of Chicago.⁴⁰ Our findings will provide insights regarding which area has the lowest accessibility of primary care and how improving spatial accessibility may impact hypertension prevalence.

Our study has several strengths. First, COMPASS is a large prospective cohort study that includes detailed information on participants' health conditions. The oversampling of disadvantaged groups enables us to investigate disease burdens in this understudied population. Second, the high-quality participant residential address and primary care location data allow us to accurately measure the spatial accessibility of primary care for each participant. Many studies on spatial accessibility were weakened by inaccuracies in data on health care providers. Many prior studies obtained data on providers from the American Medical Association/American Osteopathic Association, however not all physicians are represented in these associations.⁵ Third, our study uses measured hypertension, a specific health condition, as the major outcome and we further investigated its subclasses including uncontrolled and unaware hypertension. Most earlier studies were only able to evaluate service utilization, and thus their conclusions were vague in terms of population health. Our study overcomes this limitation and clearly shows that better spatial accessibility of primary care indeed improves population health.

Several limitations should be considered. First, the study population may not be nationally representative,

nor was it designed to be geographically representative. The oversampling of disadvantaged groups can be a strength, but also leads this study to suffer from generalizability problems. Second, the study was conducted in Chicago, an urban area. A rural area with a lower population density and fewer medical facilities may have a different pattern for spatial accessibility of primary care. Therefore, caution is needed when translating the results into rural areas. Third, this study only analyzed the baseline data from COMPASS and thus is cross-sectional in nature. We can estimate impacts on prevalence but not temporality. However, reverse causation (ie, hypertension led to worse spatial accessibility of primary care) is not plausible in this study. Fourth, we had no knowledge of how many physicians/providers were practicing in each clinic and what insurance each clinic accepted. Lack of this information might lead to underestimation of spatial accessibility of primary care and thus its associated impacts. Our findings can still speak volumes about the influence of spatial accessibility of primary care on hypertension control and awareness.

In summary, this study evaluates the association between spatial accessibility to primary care and hypertension status among largely urban minority populations in Chicago. Better spatial accessibility to primary care was associated with reduced prevalence of both uncontrolled and unaware hypertension. When stratified by neighborhood SES, the association with unaware hypertension was observed in both disadvantaged and nondisadvantaged neighborhoods, whereas the association with uncontrolled hypertension was still observed in disadvantaged neighborhoods. These findings suggest that ensuring adequate number of primary care providers can be effective to improve population health. Residents in disadvantaged neighborhoods can also benefit from interventions that increase the number of primary physicians.

ARTICLE INFORMATION

Received December 16, 2021; accepted June 8, 2022.

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Acknowledgments

The authors thank the dedicated COMPASS (Chicago Multiethnic Prevention and Surveillance Study) field staff and community partners for their support of this work. We also thank MAPSCorps for their public data on building assets in the South Side of Chicago. This research was supported by funding from the University of Chicago Comprehensive Cancer Center.

Sources of Funding

The study is supported by National Institutes of Health (NIH) Office of the Directors (OT2OD026557) and National Institute of Environmental Health Sciences (NIEHS; P30ES027792).

Disclosures

None.

Supplemental Material

Supplemental Methods

Tables S1–S4

Figure S1–S3

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